

Segregation, Political Economy, and The Longevity of Black and White Americans

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1 Abstract

Racial segregation is widely understood to advantage White Americans at the expense of Black Americans through differing neighborhood environments. This view overlooks how segregation reshapes the political economy of places through public goods, taxation, and redistributive policy that implicate the well-being of all residents. This paper links 1905–1920 birth cohorts in the 1940 full-count Census to Social Security death records ($N = 2.9$ million) and uses multiple instruments and a sibling fixed effects design to estimate the effect of segregation on the longevity of Black and of White Americans. A 10-point increase in segregation reduces *both* Black and White longevity—by 0.44 and 0.29 years, respectively—with steeper harms for less-educated Whites. I show that abundant tax revenue, welfare spending, and SNAP funding explain substantial portions of these effects for both Black and White individuals, suggesting that political-economic dimensions of segregation operate beyond neighborhoods to shape Black and White outcomes in the same direction unequally, rather than through processes of White advantage.

2 Introduction

3 Segregation and Black-White Inequalities

Racial residential segregation by race is widely understood as a visible manifestation of Black-White inequalities of various forms (Blokland, 2008) and a mechanism of persistent racial stratification (Massey et al., 1987). Black-White inequalities in the U.S. ranging from income to education to mortality are stark in size and troubling because they have remained durably large over time. Extensive scholarly research has documented how opportunities and risk are differentially distributed across Black and White neighborhoods, with risks disproportionately patterned such that Black residents live in worse off neighborhoods.

A long scholarly tradition on racial segregation has detailed the adverse outcomes of racial segregation for Blacks and Black-White inequalities (Hendi, 2024; Massey et al., 1987; Frey, 1979). Researchers have often inferred that segregation is conversely advantageous for Whites despite comparatively limited empirical attention (Brown and Homan, 2024; Quillian, 2014). The clearest evidence for this assertion can be seen in seminal work by Douglas S. Massey on racial segregation and violence: "racial segregation persists in the United States because whites benefit from it. In undermining the status and well-being of African-Americans a social problems, segregation simultaneously increases the incentives for whites to maintain the residential status quo" (Massey, 1995, p. 1229).

Zero-sum expectations about the mechanisms generating Black-White inequalities are particularly salient for race relations and form the basis of Whites' public opinions on race (Wetts and Willer, 2018; Bobo and Zubrinsky, 1996) and opposition to redistributive policy (Gilens, 1996; McGhee, 2021). Recent work has also salient for Black American's views regarding race relations as well (Warren and Valentino, 2025). In practice, these beliefs inform Whites' resistance to residential and educational integration due to a belief that segregation produces White advantage (Crowder and South, 2008; Frey, 1979; Rich, 2025; Johnson, 2019). If segregation is beneficial to Whites, this implies that efforts to desegregate – like those in the case of school segregation – would close Black-White gaps by reducing White advantages and decreasing Black disadvantage (Johnson, 2019; Massey, 1995).

Scholarly and public understanding of segregation suggests a particular relationship with Black-White inequalities; that of White gains from segregation at the expense of Black losses. Though hypotheses and conceptualizations vary across studies, there are strong expectations that segregation should be, on average, advantageous to Whites (Light and Thomas, 2019; Quillian, 2014; Massey, 1995). Despite an appearance of equivalency between inequalities and zero-sum relationships, Black-White inequalities are consistent with at least three underlying processes: (1) segregation is harmful to both groups, (2) segregation advantages Whites and disadvantages Blacks (zero-sum), (3) segregation disadvantages Blacks but does not affect Whites, yet existing papers rarely distinguish these explanations or explore the their consequences for understanding the mechanisms of inequality. These three processes are visualized in Figure 1. ¹

¹It is important to note that zero-sum relationships are used in varied ways in the literature on racial stratification

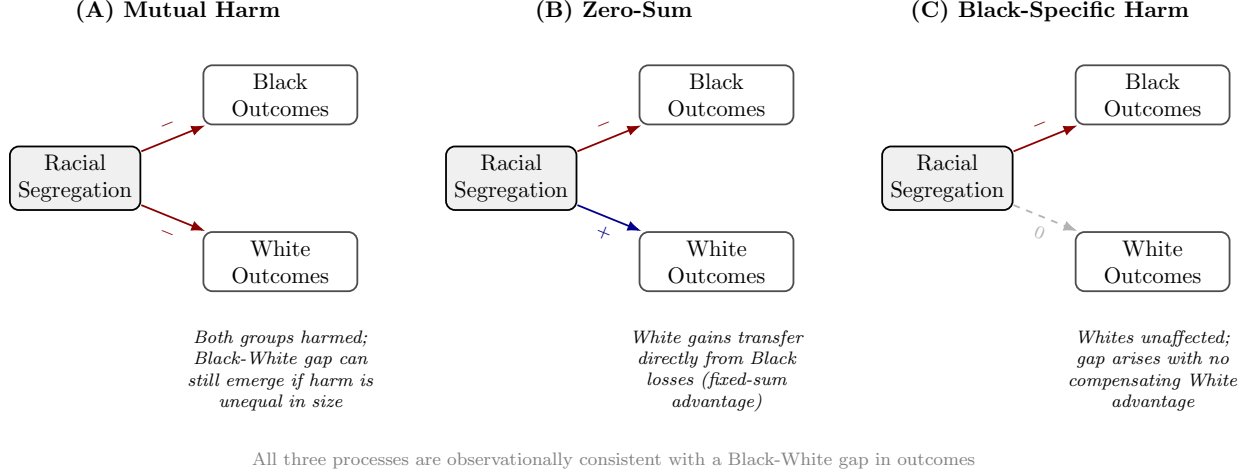


Figure 1: This figure describes three stylized theoretical processes consistent with an observed Black-White gap in outcomes. Solid red arrows denote a harmful (−) effect of segregation; solid blue arrows denote a beneficial (+) effect; the dashed gray arrow denotes a null effect. Only Panel (B) implies a strict zero-sum transfer of advantage from Black to White residents; Panels (A) and (C) generate the same qualitative inequality without any White benefit.

Simply identifying a gap between Black and White outcomes is ambiguous regarding the underlying relationships between mechanisms of racial stratification (Wrigley-Field, 2025), the sign of the coefficients, and the resulting Black-White inequalities. From a substantive perspective, the presence of an inequality must be paired with a greater understanding of what group-specific relationships generate it to fully understand the consequences of racial stratification.

The research that directly examines the effects of segregation on White outcomes has yielded inconsistent conclusions about relationships that vary in both the direction of effects (either positive or negative) and across different outcomes (e.g., crime, education, income). For instance, (Light and Thomas, 2019) find that segregation is negatively associated with White homicide victimization and positively associated with Black homicide victimization, therefore consistent with a zero-sum relationship. While Quillian (2014) finds that segregation is negatively associated with Black educational attainment and not associated with White educational attainment. On the other hand, (Cutler and Glaeser, 1997) find that segregation is negatively associated with Black educational, economic, and employment outcomes, but generally not associated or slightly negatively associated

and segregation as short hand for White advantages and Black disadvantages (McGhee, 2021). This is worth noting because the original game theoretic definition states that zero-sum relationships are ones where gains of one individual are exactly proportional to the losses of the other (von Neumann and Morgenstern, 1944). In this paper, I use zero-sum as a shorthand for relationships that create White advantage and simultaneously Black disadvantage.

Whites' outcomes.

These latter two studies are inconsistent with a zero-sum relationship, despite generating clear Black-White inequalities. These empirical patterns are inconsistent with zero-sum arguments but not readily accounted for by neighborhood-based theories of segregation. Though subtle, these distinction carry different takeaways about how changes to segregation relate to the widening or narrowing of inequalities.

4 Segregation, Neighborhoods, and Political Economy

Much of segregation scholarship focuses on the consequences of living in racially segregated neighborhoods within segregated places. Accordingly, neighborhood effects theories and evidence are often employed to connect segregation to Black-White inequalities. As demonstrated by a voluminous literature, Black residents are more likely to live in high poverty neighborhoods ([Massey et al., 1987](#); [Wilson, 1987](#); [Massey and Denton, 1993](#)), be zoned for lower quality schools ([Wodtke et al., 2023](#); [Goldsmith, 2009](#)), experience higher crime exposure ([Light and Thomas, 2019](#)), and navigate more hazardous environmental exposures ([Wodtke et al., 2022](#); [Elliott and Frickel, 2013](#)) compared to White residents. While these perspectives provide rich descriptions of how unequal distributions of various residential exposures between Black and White neighborhoods shape health and economic wellbeing, they largely overlook a burgeoning literature demonstrating that segregation shapes the political economy and of places beyond neighborhoods ([Chyn et al., 2026](#); [Ananat and Washington, 2009](#)).

Politics and policies have played a central role as mechanisms of racial stratification throughout history. In *Black Reconstruction in America* W.E.B. DuBois highlighted how racial tensions were intentionally animated by White Southern elites for the purpose of dividing working-class Black and White workers who labored under shared working conditions. ([DuBois, 1935](#), p. 700). In the Jim Crow South, segregationist policies such as literacy tests and poll taxes intended to disenfranchise Black would be voters frequently excluded many poor Whites at the same time, despite racially-motivated aims ([Perman, 2001](#), p .6). Contemporary research on politics and racial threat point toward the conclusion that public opinion about race shapes voting patterns, which in turn have downstream consequences for the policies and politics of places ([Gilens, 1996, 1995](#)).

Whites' voting patterns are a particularly salient mechanism because segregation reduces contact between racial groups and increases the salience of racial threat (Blalock, 1967; Blumer, 1958). Higher racial segregation has been shown to increase the racial resentment and conservative voting patterns of non-Black voters (Ananat and Washington, 2009). Furthermore, Chyn et al. (2026) found that racial segregation reduced support for a variety of specific policies such as support for minimum wage, redistributive spending, welfare, health spending, and opposition to police reform. These voting-related pathways link segregation to the liberalism or conservatism of policies enacted that have implications for residents, particularly those with lower-incomes, regardless of neighborhood (Metzl, 2019). To facilitate a concrete discussion of the different ways neighborhood and political-economic mechanisms of segregation affect outcomes, this article focuses on mortality to study the consequences of segregation.

The striking contemporary and historical mortality disparities between Black and White populations are a particularly salient differentiation between racial groups. In 2023, the difference in life expectancy at birth between White and Black Americans was of 4.4 years (Arias, 2020). Disparities in Black and White lifespans are notable because they have remained especially persistent in light of exceptional increases in life expectancy and demographic advancements in the last 50 years. In addition to their function as health indicators, lifespan represents an important cumulative measure of life chances (Weber, 1922) and an important axis of social inequality (Maralani and Portier, 2021; Wrigley-Field, 2024). Longevity disparities have far-reaching implications for individual social networks, collective memory, culture, political participation, and social movements that would otherwise manifest if not for shortened life (Wrigley-Field, 2025).

Scholarship in sociology and demography has established that racial segregation is a fundamental cause of health inequality (Williams and Collins, 2001) that creates worse health outcomes for Black Americans and wide Black-White health disparities, though evidence on the health of White Americans is comparatively limited and mixed. Studies of infant health illustrate this ambiguity: Austin et al. (2016) found that segregation reduced the birthweight of both Black and White infants, while Vu et al. (2024) found negative effects on the birthweight of Black infants and no effect for White infants. Evidence on mortality is similarly inconsistent; Logan and Parman (2018) found that, in certain contexts, segregation was associated with greater mortality risk for White Americans and lower risk for Black Americans, whereas Karbeah and Hacker (2023) found that

segregation increased mortality risk among both Black and White children. These divergent conclusions partly reflect methodological limitations, including reliance on death records from a single state (Logan and Parman, 2018). More fundamentally, existing evidence is largely correlational. Segregation covaries with features of places that are consequential for population health, thus correlational analyses cannot distinguish whether the observed relationships between segregation and mortality are causal.

4.1 Neighborhood Effects

According to the fundamental cause perspective, racial segregation is thought to operate through many downstream mechanisms, ranging from neighborhood exposures, socioeconomic resources, and access to health care.² As indicated in current literature on the social determinants of mortality and longevity, the mechanisms that shape health are often interlinked and compound over the life course (Breen, 2024; DiPrete and Eirich, 2006). Rather than an exhaustive review of these mechanisms, I summarize three predominant neighborhood-level mechanisms that describe the major differentiators of Black and White neighborhoods and their implications for health.

A leading pathway linking segregation to health is through increases in poverty. Segregation increases poverty concentration (Massey and Denton, 1993) and concentrates it in primarily Black neighborhoods while White neighborhoods are shielded from it (Quillian, 2012; Ananat, 2011). Poverty has been shown to have direct implications for health through its reduction of economic resources and employment opportunities. Poverty induced by segregation could also matter for health indirectly, because poverty increases individual and environmental stressors associated with worse health. Poverty is particularly consequential for the health of older adults through reductions in access to health care as well as psychosocial stress which may accelerate aging processes (Massey et al., 2018). Poverty may thus serve as both a direct input of worse health as well as compound existing racial inequalities in health at older ages, particularly after 60 years old (Geronimus et al., 2006).

The unequal concentration of crime and policing are also central mechanisms of segregation. Racial segregation distributes violent crime and aggressive policing spatially, concentrating both in

²In this section, I consider both general pathways that link segregation to health as well as those particular to older adults. This is because the Data elaborated later in this study provides a snapshot of segregation observed when an individual is 65 years old.

predominantly Black neighborhoods while insulating predominantly White ones (Bell, 2020; Light and Thomas, 2019). This spatial concentration is consequential for health because exposure to community violence elevates depressive symptoms and psychological distress (Foell et al., 2021), and aggressive policing raises the risk of injury and death (Edwards et al., 2019). Because these exposures are not symmetric across groups, segregation simultaneously *increases* the risk of homicide victimization for Black residents and *decreases* it for White residents (Light and Thomas, 2019). Crime and policing thus represent one of the clearest neighborhood channels through which segregation plausibly generates zero-sum outcomes. For older adults in particular, racialized exposure to crime operates beyond acute risk of victimization to create a culture of fear (Klinenberg, 2002) that shapes physiological and psychological health. The chronic stress of residing in high-violence environments contributes to accelerated physiological weathering between Black and White older adults (Geronimus, 1992) which shapes mortality at older ages.

Finally, racial segregation affects the quality of housing and the financial resources households can extract from them to invest in health (Breen, 2024). Segregation creates competition in housing markets that impose distinct costs on Black and White residents alike, though through different mechanisms. For Whites, Daniels (1975) documents that White renters intent on living in predominantly White neighborhoods paid a premium to segregate themselves from Black neighbors, such that the maintenance of residential separation is costly for Whites. For Black homeowners, buyers historically paid roughly a 28 percent premium to purchase homes on majority-White neighborhoods and subsequently saw those homes lose value, which eroded Black families wealth prospects Akbar et al. (2025). Because housing is the primary vehicle of wealth accumulation for most households, segregation-induced cost burdens translate into diminished resources that affect health status (Hurd and Kapteyn, 2003; Wang et al., 2026). Segregation also degrades the quality housing, particularly for Black residents, which has important implications for health and risk of disease, particularly for older adults (Evans and Chapman, 2024). Lower-quality housing in the form of poor insulation, broken features, and lack suitable amenities are associated with poor health conditions such as the onset of disability at older ages and elevated biomarkers like c-reactive protein that proxy for environmental stressors (Wang et al., 2026; Clair and Hughes, 2019).

4.2 Political Economy

Each of the neighborhood pathways described operate through the composition of residential environments themselves. At least in principle, the health consequences of neighborhoods in which Black and White residents live plausibly link segregation to zero-sum outcomes. Yet, political-economic mechanisms operate at a scale beyond neighborhoods ([Logan and Molotch, 1987](#)). Taxation, spending on health and public services ([Alesina et al., 1999](#)), and the generosity of the safety net are shaped by voters within segregated places and set at the level of counties ([de Benedictis-Kessner and Warshaw, 2020](#); [Tausanovitch and Warshaw, 2014](#)). Indeed, past work has shown that segregation increases the political polarization of places, which undermines collective political efforts and in turn, spending on public services [Trounstein \(2016\)](#). The county-level consequences carry the scope to shape the health of all residents of a jurisdiction regardless of the neighborhood they occupy.

Recent scholarship points to the importance of considering how the increasing political polarization of places and their liberal or conservative politics and policies matter for health and wellbeing ([Montez et al., 2026](#); [Torche and Rauf, 2021](#)). At the county-level, important services that shape population health are delivered, such as spending on infrastructure, health, welfare, and the level of tax revenue generated ([Manduca et al., 2025](#); [de Benedictis-Kessner and Warshaw, 2020](#)). Increasingly, high-levels of segregation have been linked to less-generous levels of public goods, which each have their own implications for health and wellbeing ([Trounstein, 2016](#); [Chyn et al., 2026](#)).

There are a variety of county-level expenditures that matter for health, but the most obvious are health and medicaid spending. Greater spending on public insurance programs have been shown to have profound effects on health and mortality; for instance, ([Goodman-Bacon, 2018](#)) found that early medicaid adoption reduced the mortality of infants substantially and non-White child mortality by 20 percent. Moreover, medicaid expansion was found to have particularly salient effects for older adult mortality, with ([Miller et al., 2021](#)) estimating that expansions tied to the Affordable Care Act reduced mortality by 9.4 percent.

The level of tax revenue generated by places is critical mechanism through which places can finance policies and programs that redistribute wealth, mitigate poverty and inequality, and invest public services. Tax revenue operates as a general mechanism that can exacerbate or mitigate

inequality across places that in turn affect life outcomes ([Rauscher and Shen, 2022](#); [O’Brien et al., 2025](#)). While health and medicaid expenditures at the county-level are directly related to health outcomes, evidence indicates that non-health spending has downstream impacts on health; [McCullough and Leider \(2016\)](#) find that higher tax revenue among other spending categories, was associated with improvements in population health and ([Mays and Smith, 2011](#)) found that increased local public health spending reduced preventable deaths by 6.9 percent. More generally, ([Lobao et al., 2012](#)) found that greater government fiscal capacity was associated with reductions in poverty and increases in job growth which directly shape health outcomes.

In addition to tax revenue and health expenditures nutrition and welfare programs have direct and indirect effects on health; for instance spending on supplemental nutrition assistance (SNAP) has been shown to have direct effects on mortality; for instance, ([Heflin et al., 2019](#)) found that SNAP participation reduced all-cause mortality among adults ages 40–64 by 1–2 percentage points. Evidence from the county-level rollout of the original Food Stamp Program points to similar conclusions over the long run, with childhood access to the program increasing adult longevity by 0.4 percentage points ([Bailey et al., 2024](#)) and reducing the incidence of metabolic syndrome in adulthood ([Hoynes et al., 2016](#)). Nutrition assistance thus shapes mortality both directly and through the accumulation of health over the life course.

Beyond nutrition assistance, broader welfare spending encompasses the range of transfers and services that counties provide to their poorest residents that vary considerably across counties ([Bruch et al., 2018](#)). This spending has been shown to have direct implications for mortality; for instance, ([Fishback et al., 2007](#)) found that New Deal relief spending during the Great Depression reduced infant mortality, suicides, and deaths from infectious disease in a panel of 114 cities. Additionally, cash transfers have been shown to have lasting consequences for longevity; for instance, ([Aizer et al., 2016](#)) found that the sons of mothers accepted to the Mothers’ Pension program, the first cash welfare program in the U.S., lived approximately one year longer than the sons of rejected applicants. Contemporary evidence is more mixed, with ([Muennig et al., 2020](#)) estimating that state supplements to the Earned Income Tax Credit improved ten-year survival among adults, while ([Wyndham-Douds and Cowan, 2024](#)) found that cash assistance had no effect on birth outcomes.

Despite a burgeoning evidence base demonstrating the importance of the political economic effects of segregation, scholarship in sociology rarely tests the extent to which these mechanisms

explain the effects of segregation on the outcomes to the neighborhood-level mechanisms outlined. Determining whether the effects of segregation are more consistent with micro or macro-level mechanisms is of central importance (Sharkey and Faber, 2014). This is especially relevant in the context of recent findings that document the rising spatial scale of segregation that reflects the separation of racial groups between places (Carlson et al., 2026; Lichter et al., 2015). While macro-segregation currently characterizes about half of the metropolitan areas in the U.S. in 2020 (Carlson et al., 2026), the increasing overlap between racial composition and macro-level geographies demands a greater understanding of how political dimensions of segregation affect individual outcomes.

5 The Present Study

Thus far, the literature on segregation encompasses several mechanisms that plausibly affect mortality, though macro-level mechanisms remain at present underexplored. Neighborhood effects mechanisms broadly suggest that segregation should produce health advantages for Whites and disadvantages for Blacks, which is consistent with a zero-sum understanding of the consequences of segregation. On the other hand, political-economic mechanisms suggest broader place-level consequences that affect both Black and White residents similarly.

The present study is organized around two empirical goals. First, I estimate the causal effect of segregation on the longevity of Black and White Americans using a set of instrumental variable strategies that induce greater geographic sorting by virtue of quasi-random spatial boundaries and a sibling fixed effects design leveraging within-family variation in neighborhood attainment. In models stratified by race, I examine how racial segregation affects Black and White longevity separately to learn about the specific relationships that underpin Black-White inequalities. This approach to analyzing mortality disparities is transparent because it provides a quantification of *how* segregation affects Black and White longevity, rather than documenting observed disparities between groups. It is important to highlight that only in a scenario where the sign of the estimate on segregation is negative for Blacks and positive for Whites, would the relationship between segregation and longevity be considered zero-sum. From these analyses, I am able to recover Black-White differences in longevity from the estimates and differentiate between the underlying processes that generate inequalities.

Second, I test to the extent to which several measures of county-level spending on public goods and tax revenue explain the observed relationships between segregation and longevity. These mechanisms are (1) county tax revenue, (2) medicaid spending and health expenditures, (3) SNAP spending, and (4) welfare and cash assistance spending. While far from comprehensive, these different measures of county policy support provide a direct assessment of the level of policy liberalism with constructs and measures analogous to those used at the state-level [Montez et al. \(2020\)](#) and have strong theoretical links to individual health outcomes. This effort also builds upon the analyses in [\(Chyn et al., 2026\)](#) demonstrating that racial segregation reduces support for progressive policies by examining how much observed public spending outcomes explain the effects of segregation on health.

6 Data

This study links six sources of data together to study the relationship between racial segregation and mortality: (1) individual-level data from the 1940 Census, which contains demographic information (race, gender, education, birth year); (2) linked mortality records from the CenSoc Numident data file, which contain information on year of death, place of death, and place of birth ([Breen et al., 2023](#)); (3) county-level summaries of the racial composition and segregation of places from the decennial census (1980-2000) ([Ruggles et al., 2023](#)); (4) data on historical railroad line shapefiles established before 1911 ([Atack, 2016](#)), which are used to construct the first set of instruments; (5) county-level data on the arrangement of rivers, streams, and waterways from the TIGER/Line linear hydrography files ([U.S. Census Bureau, 2023](#)), which are used to construct the second set of instruments; and (6) government finance data from the Government Finance Database ([Pierson et al., 2015](#)), which are used to calculate county-level expenditures and tax revenues to test political-economic mechanisms.

The data cover deaths occurring between 1988 and 2005 among members of the 1905–1920 birth cohorts. I link these sources such that each observation represents an individual and includes their demographic characteristics measured in 1940, county of birth, year and county of residence at death, and the segregation, contextual, instrument, and policy measures of the county in which they died.

6.1 Linked 1940 Census–Numident Mortality Data

The empirical analyses in this paper are anchored by individual-level demographic data records from the 1940 Full Count Census (Ruggles et al., 2023). To construct my analytic sample, I identify individuals from the 1905-1920 birth cohorts who were 20-35 in 1940. These individuals are then linked to Social Security Administration (SSA) mortality records between 1988 and 2005 provided by the Berkeley CenSoc Numident file (Breen et al., 2023). These data offer unparalleled individual-level data on mortality that are geolocated to contemporary residence at death and which are representative of the population living in the United States in 1940. The geographic specificity and rich covariates make these data ideal for analyzing spatial inequalities in mortality. CenSoc Numident records cover deaths occurring between 1988 and 2005, allowing for reliable estimates of longevity conditional on an individual surviving to age 65 (Goldstein et al., 2023). The interpretation of longevity conditional on surviving to 65 arises because of the death window that is reliably covered by the Numident sample (Breen et al., 2023, p. 35) and is the interpretation used in Breen (2024).. The implications of this condition for the results are addressed directly in Section 8.4.

I focus on the 1905-1920 birth cohorts for two reasons. First, their birth years and the end year of the mortality observation window allow for plausibly complete observation of mortality that is not biased by right censorship (i.e. missing the oldest old). In other words, those born in 1905 would be 100 years old in 2005, making it reasonable to assume that few individuals remain alive. Even for those individuals who live very long, using doubly truncated data will result in estimates biased toward zero (Breen, 2024), and thus produce estimates of the effect of segregation on longevity that are likely conservative.

Second, restricting to ages 20-35³ allows for confidence that individual-level confounders to longevity, such as education status, are observed. By conditioning on those 20 years or older, it is reasonable to assume that individuals have completed the modal level of formal education by the time that they appear in the sample.

Despite the strengths of the Numident data, there are limitations worth noting. The first is

³Halpern-Manners et al. (2020) finds that the mean number of years of education of decedents in the 1940 Census is 10.4 years (1525). If we assume a normative time of starting school at 6, individuals in the sample reach the average at 16.4 years, which is well before our age cutoff.

double truncation, where data are unobserved on both the left and right side of the death window (e.g., before 1988 and after 2005). [Goldstein et al. \(2023\)](#) show that when not accounted for, doubly truncated mortality observation windows result in biased estimates of longevity disparities that shrink toward 0 and higher mean observed death age in samples.⁴ This means that those born in 1905 are observed between the ages of 83-100 years old and those born in 1920 are observed between the ages of 68-85 years old. Birth year fixed effects help to account for some of this bias ([Goldstein et al., 2023](#)), where individuals of varying birth cohorts may have differing numbers of years covered by the death window. However, this cannot be fully accounted for in analyses. Fortunately, a quantification of the bias is available using a Gompertz maximum likelihood technique ([Goldstein et al., 2023](#); [Breen, 2024](#)), which I implement and return to in Section 8.4.

The second limitation concerns race-specific representativeness. The CenSoc Numident file links 1940 Census data to the public release of the National Archives Record Administration “NARA Numident” file using the Abramitzky, Boustan, and Eriksson (ABE) exact matching algorithm ([Abramitzky et al., 2012](#)). For the Numident sample, 22% of the cases in the 1940 Census are matched to mortality records, which is concordant with other historical census matching efforts ([Breen et al., 2023](#), p. 7). A natural question is whether the differences in match rates are sufficient to produce bias in results when studying mortality with the Numident linked data. For Black men and women, comparisons with the 1940 Census indicate that the Numident file slightly underrepresents these groups, at -1.8 and -2.4 percent differences, respectively ([Breen et al., 2023](#)).⁵ ([Breen et al., 2023](#)) argue that bias due to representivity difference is modest when analyses are stratified by race. Nevertheless, [Osborne \(2022\)](#) provide a set of post-stratification weights that re-weight CenSoc Numident counts to period-specific distributions of deaths to adjust for lingering selection from matching differences. To address this, I re-run models using these weights are included in appendix Table Axx.

Additionally, analyses using sibling information, described in the Methods section, draw on a subsample of the analytic sample in which siblings can be identified within 1940 households. Siblings are identified using the ABE algorithm according to the procedure in ([Breen, 2024](#)), but

⁴The high average age at death for both Black and White individuals can be seen in Table 1 transparently and this is an artifact of truncation.

⁵This matching procedure may be particularly selective for unmarried women, because of marriage-related name changes over the life course. Fortunately the Numident data contain parents’ last names that can be used to link women who have undergone name change.

produces a much smaller sample size than the full Numident data. The differences in individual characteristics of these samples are discussed in later sections.

6.2 Segregation and County Characteristics at Death

I examine exposure to segregation in mid-to-late life, obtained and calculated based on the county individuals resided in at death. Examining the county at death offers an empirical understanding of the effects of segregation that occurred after the Great Migration and the passage of Civil Rights Act that outlawed legal segregation. Because much of the contemporary segregation scholarship examines the consequences of de jure segregation, this time period aligns more consistently with these theories. Scholars recognize that the geographic scope and structure of segregation after the Great Migration are understood to be distinct from the segregation that occurred before, predominately in the South ([Leibbrand et al., 2020](#); [Grigoryeva and Ruef, 2015](#)).

There are of course limitations to using one’s county at death. It is reasonable to believe that other exposures that occur throughout the life course, such as the place characteristics of where individuals grew, up affect their health and mortality long after. To address these potential exposures, I include birth county fixed effects to absorb early life characteristics. Yet, it is nevertheless likely that segregation in the county at death net of birth county fixed effects still miss intervening exposures between birth and death matter for health. Patterns of neighborhood attainment and residential mobility over the life course show that the modal neighborhood transitions are to places with similar characteristics ([Sampson and Sharkey, 2008](#)). Moreover, across the life-course, migration propensity declines with age ([Bernard et al., 2014](#)): individuals are much less likely to migrate after age 40 and, when they do, it is usually short distances ([Molloy et al., 2011](#); [Zimran, 2022](#)). For these reasons, this paper’s focus on segregation in one’s the county of death plausibly represents a period of exposure encompassing mid-to-late life, but this is a limitation.

To measure segregation, I use the dissimilarity index (D) ([Duncan and Duncan, 1955](#)). The dissimilarity index measures the evenness of the distribution of Black and White residents within a county, ranging from 0-100, and is interpreted as the share of Black or White individuals that would have to move to a different neighborhood for the racial composition of neighborhoods to be the same as the racial composition of the county. The dissimilarity index is widely used in prior work ([Hendi, 2024](#); [Light and Thomas, 2019](#); [Massey and Denton, 2003](#); [Cutler and Glaeser,](#)

1997) because it is straightforward to calculate, concordant with past empirical evidence, and has a concrete interpretation. The dissimilarity index for the county an individual resided in at the time of their death is calculated formally in Equation (1):

$$D = \frac{1}{2} \sum_i^N \left| \frac{b_i}{B} - \frac{w_i}{W} \right| \times 100 \quad (1)$$

Where N is the number of counties, b_i and w_i are the proportion of Black and White residents in the i th census tract and B and W are the proportion of Black and White residents in the county. The dissimilarity scores are calculated from decennial Censuses 1980, 1990 and 2000, which correspond to the year of death observed for sample members.⁶

6.3 Instrument Data: Railroads and Waterways

I construct the instruments used in analyses from two sources of geospatial data. First, I use Jeremy Atack’s historical GIS database of U.S. railroads, which digitizes the network of rail lines placed in operation between 1826 and 1911 (Atack, 2016). Retaining all segments in operation by 1911, I overlay the historical rail network on county boundaries and construct the Railroad Division Index (RDI) developed by Ananat (2011), which is described in detail in the Methods section. I also compute an indicator for any pre-1911 railroad, total track length, and track density (kilometers of track per square kilometer of land area).

Second, I measure the arrangement of rivers and streams within counties using the U.S. Census Bureau’s TIGER/Line linear hydrography shapefiles (U.S. Census Bureau, 2023), in the spirit of the topographical instruments introduced by Cutler and Glaeser (1997). From these files, I retain natural stream and river features and exclude man-made canals, ditches, and artificial connectors, which are plausibly endogenous to development.

⁶Although widely used and the interpretation convenient the D is well known to suffer from various index biases, outlined in Fossett (2017) and others (Winship, 1977; Carrington and Troske, 1997), particularly as they relate to temporal comparisons and variability in the population composition and relative group size of counties. Accordingly, in supplemental analyses, I replicate the main results with a bias-adjusted D index that adjusts for this bias (Fossett, 2017). In contrast to D , the bias-adjusted D measure is more complex to estimate and the interpretation less concrete (Fossett, 2017, p. 233). For these reasons, I use the conventional dissimilarity index for main analyses.

6.4 Political-Economy Mechanism Data

To measure county-level political-economic mechanisms, counties are linked to government finance records from the Government Finance Database (GFD), a compilation of Census of Governments and Annual Survey of Governments finance data (Pierson et al., 2015). From the GFD, I extract the finances of county governments and construct measures of total tax revenue (including property and individual income tax components), health and hospital expenditures, Medicaid vendor payments, education spending, and direct welfare and cash assistance spending. Because government units are surveyed at varying frequencies, finance records are summed within each county-year and then averaged within the decade windows corresponding to the death decades observed in the sample (1980–1989, 1990–1999, and 2000–2005), so that each individual is assigned the policy environment of the decade in which they died. Each measure is expressed per 1,000 county residents using decennial census population denominators. *[[Dollars are currently nominal decade averages (Code/04_policy_data.R); decide whether to deflate to constant dollars and, if so, state the base year.]]*

6.5 Analytic Sample

My main analytic sample consists of ($N = 2,412,213$) individuals residing in X counties at their time of death over three decades from 1980–2000. The resulting counties are filtered from all counties in the U.S. such that they have a Black population greater than 0 and coverage of the RDI instrument ⁷. *[[Update the filtering criteria, footnote, and Table A1 (FigTab/Filtering_table.tex): the sample previously also conditioned on non-missing 1967 Census of Governments finance measures used for the government-fragmentation instrument, which is now removed. Restate the filtering steps for the railroad and river instruments (e.g., counties with valid RDI/hydrography measures) and report the new stepwise Ns.]]*

The sibling sample, described in the Methods section, relies on a subset of the sample containing information on siblings in the CenSoc Numident

Table 1 displays the individual and county characteristics of the 1905–1920 birth cohorts, separately for Black and White decedents, across the full data, the IV sample, and the flexible sibling sample.

Table 1 demonstrates that there is striking consistency in the observable characteristics of

⁷The data filtering procedure and stepwise sample sizes are included in table A1 in the Appendix.

Table 1: Representativeness of Analytic Samples by Race

Variable	Full Data		IV Sample		Sibling Sample	
	Black	White	Black	White	Black	White
Age at Death	81.55 (6.02)	82.05 (5.66)	81.55 (6.02)	82.05 (5.66)	80.82 (5.81)	81.53 (5.48)
Male	0.42 (0.49)	0.44 (0.50)	0.42 (0.49)	0.44 (0.50)	0.44 (0.50)	0.47 (0.50)
Education (Years)	7.57 (3.50)	10.44 (3.02)	7.57 (3.50)	10.44 (3.02)	8.14 (3.48)	10.47 (2.91)
Married (In 1940)	0.55 (0.50)	0.49 (0.50)	0.55 (0.50)	0.49 (0.50)	0.44 (0.50)	0.39 (0.49)
Migrated (Birth to Death County)	0.27 (0.44)	0.30 (0.46)	0.27 (0.44)	0.30 (0.46)	0.30 (0.46)	0.32 (0.47)
Reside in South at Death	0.61 (0.49)	0.30 (0.46)	0.61 (0.49)	0.30 (0.46)	0.59 (0.49)	0.27 (0.44)
County D at Death	0.57 (0.20)	0.53 (0.17)	0.57 (0.20)	0.53 (0.17)	0.56 (0.19)	0.52 (0.17)
County Population (1000s)	1,003.44 (1,777.88)	759.70 (1,469.92)	1,003.67 (1,778.16)	760.17 (1,470.77)	1,007.71 (1,811.77)	721.53 (1,406.21)
County Pr. Black	0.26 (0.17)	0.10 (0.11)	0.26 (0.17)	0.10 (0.11)	0.25 (0.17)	0.09 (0.11)
N	125,122	2,290,150	125,075	2,287,138	12,314	414,884

Cell entries are unweighted means with standard deviations in parentheses. The IV sample is restricted to counties with non-missing railroad density index and rail density data and to individuals with non-missing analytic covariates. The sibling sample is the flexible sibling-group match on the full data and is not restricted to counties with non-missing railroad density index data, so it is not a subset of the IV sample.

Black and White individuals across samples, which provides support for the external validity of comparisons of estimates across samples.

[[Model-paper move worth copying: after Table 1, add 2–3 sentences narrating the key comparisons (as Donahue & Torrats-Espinosa do for their Tables 1 and 3), e.g., how in-sample individuals compare to out-of-sample cohort members on race, education, and region.]]

7 Empirical Strategy

Next, I turn to the main analysis plan which estimates the causal effect of racial segregation on the longevity of Black and White Americans using a set of instrumental variables (IV) and a sibling fixed effects models. The logic of these instruments utilize quasi-random natural and historical variation in physical barriers that divide up counties to instrument strategy instrument for racial segregation. Greater spatial division of a county, thus facilitates more segregation within places⁸. Instrumental variable designs are powerful because they offer one strategy to identify the effects of segregation in a setting where there are many individual selection factors that may shape where individuals live, which in turn shapes their health. They also offer a high degree of geographic representivity in the sample, but rely on strong assumptions that are difficult to test. My approach in this paper is to triangulate the plausibly causal effect of segregation on longevity using a variety of complementary strategies anchored by the IV estimates. I first describe the necessary IV assumptions and evidence in support of their validity, then I present a complementary instrument strategy, and finally an alternative sibling fixed effects design.

IV designs rely on three core assumptions. First, relevance requires that the instruments strongly predict segregation (Angrist and Kolesár, 2021). Second, the exclusion restriction requires that instruments affect longevity only through segregation. It is important to recall that the instruments used in this setting measure the *configuration* of barriers within a county rather than any quality of the barriers themselves or of the places they divide. I take the threats to this assumption seriously and address them at length below, testing whether the RDI is directly associated with education, income, and pollution exposure, and complementing the railroad strategy

⁸Similar instrumental variable designs have been used to examine the effect of segregation on educational outcomes (Cutler and Glaeser, 1997; Quillian, 2014), poverty and life expectancy (Ananat, 2011; Hendi, 2024), political composition (Ananat and Washington, 2009), and public spending (Trounstine, 2016).

with natural boundaries that are not the product of human placement.⁹ Third, monotonicity requires that the instruments push segregation in the same direction for all units. While a direct test of this assumption is not possible, I examine demographic patterns that would be consistent with monotonicity and discuss them in Appendix B. Conditional on these assumptions, IV estimates are interpreted as the local average treatment effect (LATE) (Imbens and Angrist, 1994; Mogstad et al., 2021), or the treatment effect among those living in counties where segregation increased because of greater physical division.

The preferred instrumental variable strategy is the railroad division index (RDI) proposed by (Ananat, 2011). This instrument is constructed for counties¹⁰ using historical railroad shapefiles sourced from (Atack et al., 1994). The RDI measures the extent to which counties are divided up by intersections historical railroad lines that were established in the early 20th Century. The RDI calculates the number of intersections that a county has with rail lines in a standardized index that ranges from 0 to 1, with higher values indicating more division. In addition to rail lines, I also instrument with the density of railroad tracks per square kilometer of the county. The calculation of the RDI is described as follows:

$$RDI = 1 - \sum_i (a_i/A)^2 \quad (2)$$

Where a_i is the area of each subpolygon created by cutting the county with the rail network and A is the total county area. The index equals 0 for counties left undivided by railroads and approaches 1 as rail lines subdivide a county into many small pieces, capturing the physical partitioning of places that facilitated racially divided neighborhood boundaries.

The intuition behind the RDI is that historical railroads provided natural barriers within places

⁹Because there are more instruments than endogenous regressors, a natural test of the validity of many instruments are overidentification tests. These tests are commonly interpreted as testing the “validity” of the instruments, however overidentification tests must first assume that one instrument is valid (e.g., in the just-identified case) (Felton and Stewart, 2024), which is not informative for assessing validity. A more informative interpretation from Mogstad et al. (2021) is that rejection of overidentification tests suggests that there is heterogeneity in treatment effects across subpopulations affected by the different instruments, which is not in reference to validity. Additionally, overidentification tests assume constant treatment effects, which is unreasonable in a setting where the IV estimate is a local average treatment effect. As expected, the instruments generally fail to pass these tests given the large sample size of my data, where observations are in the millions. An alternative to overidentification tests is placebo analyses that require a researcher find an additional instrument that shares all the same confounders as the original instrument but is thought to not affect the treatment (Felton and Stewart, 2024). While informative, the shared confounders assumption of placebo instruments is also quite strong and the search for a candidate placebo instrument faces the same challenges as the search for a credible instrument.

¹⁰This has been used prior in (Ananat and Washington, 2009; Chyn et al., 2026) at the metro-level.

that populations could cluster around, thus creating higher levels of baseline segregation. (Ananat, 2011) also demonstrates that historical White segregationist efforts often utilized natural barriers such as rail lines to establish communities and neighborhoods, because these barriers are naturally difficult to navigate.

There are two leading issues that may undermine exclusion. First, in its original usage (Ananat, 2011), the RDI is constructed for Northern metropolitan areas, because the majority of African Americans lived almost entirely in the South before the 1900s when the railroads were laid. Therefore only in non-Southern cities can railroad configuration be treated as exogenous to race and thought to induce segregation by Black inflows from the great migration. This the logic of exclusion rests on this restriction but limits the geographic generalizability of estimates. To overcome this, I examine the consistency of results in models stratified by those living in the South versus the non-South and adjust for residence in the South in all models and these results are quantitatively similar (Appendix table Axx).

Second, it is important to note that while railroad division does not directly relate to railroad *exposure* per se, they are inherently related. The exclusion restriction hinges on the assumption that early 20th Century railroads do not affect mortality directly. On one hand, railroads could affect mortality by generating financial prosperity in areas surrounding tracks which could promote better health. (Black et al., 2015), using railroad proximity to instrument for migration, they show that railroads had little association with economic outcomes for a non-selective group of Southern-born White older adults, indicating that the long-term health impacts of railroad induced prosperity are likely minimal. To test this possibility directly, I regress education and income of 1905-1920 birth cohorts in 1940 on the RDI to test whether railroad division is directly related to socioeconomic outcomes. I conduct this procedure for those that did not migrate during their lifetime to capture the potential impact of the RDI for those exposed continuously to the same place as well as the full sample to capture the relationship between exposure to the RDI in one's county at death (Appendix tables Axx). These tests indicate that the RDI is not significantly associated with the educational attainment of Blacks or Whites, regardless of sample. However, the RDI is associated actually slightly negatively associated income for Blacks and Whites at a similar magnitude ($-.04$). However it is important to note the small magnitude of this association. The RDI ranges from 0-1, indicating that a one point increase in railroad division (RDI/100) would translate to approximately

a 0.004% decrease in earnings in 1940. While these estimates are significantly negative, they are quite insignificant substantively. ¹¹

On the other hand, railroads could also result in increased hazards such as pollution and or negative environmental externalities that harm health (Dockery et al., 1993). In particular, small particulate matter in the air has resulting from exhaust and emissions is associated with poor health outcomes [cites]. To test the extent to which the RDI could directly affect health through pollution, I examine the relationship between the RDI and small Particulate Matter (PM2.5) concentration $PM2.5/m^3$ and railroad division in U.S. counties in 1999 and 2000. ¹² PM2.5 measures the concentration of tiny dust and soot particles in the air in micrograms per cubic meter (μ/m^3), with an safe range concentration ranging between 0-9 within an area. Tests of the association between RDI and PM2.5 concentration are positively and significantly associated, but the magnitude is very small. Going from a county that has no railroad division to one that is completely divided by rail lines represents a difference in PM2.5 concentration of .356 that, while significantly associated, is quite small in magnitude, which suggests that the direct effects on mortality are minimal (Appendix Axx).

Despite evidence presented against exclusion violations owing to direct effects of the RDI on health and the expansion of the to Southern states, it is nevertheless possible that there is still bias present. The second set of instruments employed to complement this strategy utilize the density of natural rivers and streams within a county ¹³. For each county, I compute the number of distinct named rivers and along with total stream length and stream density (Trounstein, 2016) ¹⁴.

The logic of using natural streams and rivers to instrument for segregation relies on the quasi-randomness of these natural boundaries. Like the RDI, rivers and streams divide up areas in ways that shape population settlement patterns around them, which leads to greater geographic clustering that is plausibly unrelated to health. While railroads and highways have been used historically

¹¹It is important to highlight that the regression results for income and education for those who did not migrate during their lifetime (i.e., those that were can confidently attribute their RDI value to birth place as well as death place) are highly similar to the full sample results, suggesting that whether we capture RDI exposure early or later in life, any impacts do not appear to compound over time in ways that exacerbate exclusion restriction violations.

¹²Prior to the 1990s, the Environmental Protection Agency did not collect data on PM2.5 levels. Modeled estimates are available for earlier periods, but may be subject to bias.

¹³Each of these instruments relying on spatial division will be correlated to some degree, as demonstrated in the appendix figure (Axx), however the correlation between instruments is relatively weak. Across all pairwise correlations the instruments range from an R^2 value of x to xx . These relatively low values indicate that each of these instruments is picking up on a different dimension of spatial division in an area rather than substituting for another measure.

¹⁴Man made rivers and streams are omitted.

to segregate, those boundaries are man-made, which raises the concern that they could be endogenous to segregation. Natural boundaries are not subject to these sources of bias because difficult for humans to manipulate. Rivers and streams also overcome the issues posed by generalizing the RDI to the South, because they do not rely on any particular history of settlement. Despite the river and stream density overcoming the limitations of the RDI, they may still carry implications for health. One possibility is that more rivers and streams are correlated with greater availability of green spaces and natural areas that carry health benefits, thus violating the exclusion restriction.

Thus far, different instrument strategies carry the shared concern that there could be some features of their physical and geographic nature that could be affect health. To complement these IV approaches, I utilize a sibling fixed effects identification strategy ¹⁵. Sibling designs identify effects by controlling for unobserved heterogeneity at the family-level by differencing out shared factors common among siblings (e.g., family wealth, health endowments, genetic factors, etc.). After controlling for individual demographic characteristics that vary across siblings, we identify the effect of segregation on longevity from differences in siblings residential environments.

The sibling fixed effects strategy is advantageous in this study context, because it relies on different assumptions and sources of identifying variation to establish a causal relationship between segregation and longevity. While sibling fixed effects require more modest assumptions, they also carry important considerations regarding external validity, due to their reliance on a subset of matched sibling pairs and discordance in residential environments among siblings. Fortunately, observable characteristics of the sibling sample track the full sample and IV samples closely (as shown in Table 1), providing confidence in their representivity compared to the broader population.

Each of these designs rely on different identifying variation and assumptions for identifying causal effects. Because, for example, sibling designs do not require the same exclusion restriction assumptions as the instruments, then consistent estimates across designs suggest that exclusion violations do not drive results, which allows for the triangulation of the effect of segregation on longevity.

¹⁵Sibling fixed effects strategies have been used in prior literature to estimate the effects of education on mortality (Halpern-Manners et al., 2020) and the effects of neighborhoods on intergenerational mobility (Chetty and Hendren, 2018)

7.1 Estimation Strategy

The estimation strategy in this paper revolves around three sets of models stratified by race: OLS regression models, Instrumental Variable regression models, and sibling fixed effects models. First, I estimate a set of OLS regression models of the form:

$$(OLS) \quad Y_i = \beta_0 + \beta Segregation_i + \boldsymbol{\beta} X_i + \gamma_{b(i)} + \lambda_{c(i)} + \epsilon_i \quad (3)$$

Where $\beta segregation$ is the dissimilarity in an individual's county at death. In all models, segregation is expressed on a scale from 0 – 100 so that the estimate corresponds to each unit increase in dissimilarity rather than the contrast of a perfectly integrated or segregated city. $\boldsymbol{\beta}$ is a vector of regression coefficients and X is a vector of individual-level control variables and γ and λ are fixed effects for individual birth year and birth county

Second, I estimate a set of Instrumental Variable regressions via two-stage least squares estimation according to the following form:

$$(First-Stage) \quad \beta Segregation_i = instrument \in \begin{cases} RDI + Rail \ Length/km^2, \\ N \ Waterways + Waterway \ Length/km^2 \end{cases} \quad (4)$$

$$(IV) \quad Y_i = \beta_0 + \beta Segregation_i + \boldsymbol{\beta} X_i + \gamma_{b(i)} + \lambda_{c(i)} + \epsilon_i$$

Where the form of the model is the same as OLS, however, now segregation has been instrumented by one of the instrument sets. In contrast to regression models estimated by OLS, the coefficient on segregation changes to the effect of segregation on the longevity of compliers: those whose counties became more segregated in response to a denser configuration of physical barriers.

Third, I estimate the sibling fixed effects models on the subsample of sibling data using the same regression specification as equation (3), however, in this model an identifier that groups individuals into siblings within families is added. The models are estimated as follows:

$$\text{Sibling FE } Y_{ij} = \beta_0 + \beta \text{Segregation}_i + \beta X_i + \gamma_{b(i)} + \lambda_{c(i)} \sigma_{d(j)} + \epsilon_i \quad (5)$$

Where $\sigma_{d(j)}$ is a family fixed effect and the indexing of the model has been adapted to index individuals i within families j .

In all models, I control for a rich set of individual and contextual covariates to adjust for any residual confounding. In fully specified models, I control for sex, education, marital status, employment status, whether an individual migrated between birth and death, whether they reside in the South, urban-rural fixed effects ¹⁶, birth year and state fixed effects, and occupation fixed effects.

It is important to note the multi-level structure of: segregation is measured at the county level, while individuals are clustered within counties. Because the regressor of interest is constant within cluster and errors are plausibly correlated within cluster, default standard errors are potentially severely downward-biased (Cameron and Miller, 2015). In all models I therefore report cluster-robust standard errors at the county level. This addresses the same source of dependence that a random-intercept multilevel model would absorb, but does so without requiring that the within-cluster error covariance structure be correctly specified, which is the condition under which the FGLS/multilevel estimator’s efficiency and inferential validity hold (Cameron and Miller, 2015). Since my interest is in the county-level association rather than in variance decomposition, the clustering solution is preferred because it imposes fewer assumptions on the error process.

Finally, while the conceptual interest of this paper is to understand the *specific* relationships and mechanisms that underlie the relationship between segregation and longevity disparities between Black and White Americans, the empirical goal is to estimate $\beta_{\text{segregation}}$ separately by race. This is done so that the race-specific effects of segregation can be empirically separated from the gap between groups. Once the race-specific effects are known, the Black-White longevity gap can be recovered from the difference between the estimates of segregation on longevity for Black and

¹⁶Between 1993 and 2003, Rural-Urban Continuum codes changed from their definitions between 1983-1993. For 1983-93 codes, metro counties were classified using a scheme ranging from 0-3, with counties that belonged to a metro area with a population of 1 million or more were categorized into Central and Fringe counties. In 2003, this scheme was revised so that counties were classified based only by population, with Fringe and Central collapsed into one category. Accordingly, to increase comparability, I collapse codes 0 and 1 in the 1983 and 1993 continuum code schemes into one category so that they are harmonized 1-3 scheme used in 2003.

White regressions.

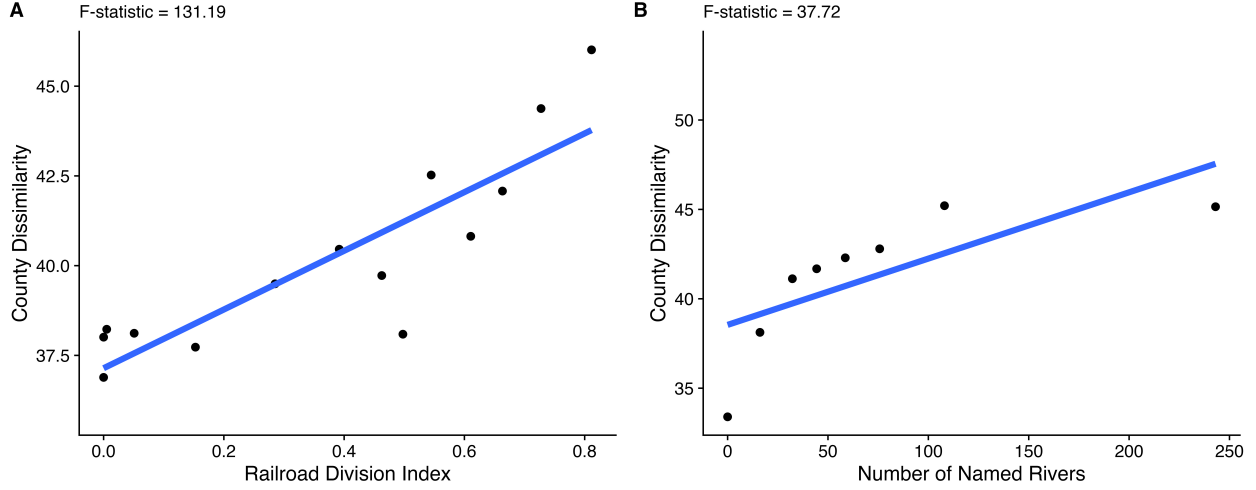
8 Results

I present the results in the order that the empirical strategy sets up. I begin with the relevance of the instruments, because the credibility of the instruments rest on this assumption first. I then report the OLS associations between segregation and longevity separately for Black and White individuals, which establish the sign and magnitude of the relationship in the dataset. I then turn to the instrumental variable estimates and sibling fixed effects estimates to triangulate the effects of segregation on longevity across empirical strategies. Throughout, I estimate all models separately by race and express the estimates as the effect of a 10-point increase in the dissimilarity index.

I begin by testing the relevance assumption, which requires that the instruments strongly predict segregation. Figure 2 displays the first-stage relationship between each instrument and county dissimilarity, residualized on the remaining instrument in each strategy. Panel A shows that counties more finely divided by the pre-1911 railroad network are substantially more segregated: a county at the top of the RDI distribution is roughly 9–13 points more segregated than an undivided county, and the first-stage F -statistic of 78.0 far exceeds conventional thresholds for instrument strength. Panel B shows the analogous relationship for the natural-boundary strategy, where counties containing more named rivers are more segregated, with a first-stage F -statistic of 34.0. In the race-stratified first stages reported in Tables 3 and 4, the RDI coefficient is 8.9 in the Black sample and 7.5 in the White sample, and both are precisely estimated. The instruments are therefore strong predictors of segregation in both samples, and the two strategies recover first stages of similar magnitude despite drawing on entirely different sources of spatial division.

8.1 Main Estimates of Segregation on Longevity

Table 3 reports the OLS association between segregation and longevity, estimated separately for Black and White decedents with birth year, birth state, and urban-rural fixed effects, and with the addition of individual controls and occupation fixed effects in the even-numbered columns. The association is negative for both groups and is essentially unchanged by the inclusion of controls. Among Black individuals, a 10-point increase in dissimilarity is associated with 0.71 fewer years of



This figure displays the association between each instrument and segregation. Segregation is measured by the index of dissimilarity that measures how evenly distributed Black and White residents are within a county. Panel A displays the Railroad Division Index first stage, adjusting for railroad density. Panel B displays the named rivers first stage, adjusting for stream density. Dots represent means of bins at each level of the instrument. The blue line corresponds to the fitted OLS regression line of segregation on the instrument.

Figure 2: First-Stage Relationships Between the Instruments and Segregation

life; among White individuals, the same increase is associated with 0.65 fewer years. The stability of these coefficients across specifications is notable: adding sex, education, marital status, migration status, and occupation fixed effects moves the Black estimate not at all and the White estimate by less than a hundredth of a year, which suggests that the association is not an artifact of the individual-level composition of segregated counties.

Because the samples used by each identification strategy differ, the details of which are described in the Appendix, I re-estimate the OLS association on each of the four estimation samples that the subsequent models use. Figure 3 displays these estimates. The point of this figure is that the OLS association is remarkably insensitive to which sample it is estimated on. Across the RDI, rivers, flexible sibling, and exact sibling samples, the Black estimate ranges from -0.73 to -0.61 years and the White estimate is essentially fixed at -0.65 years per 10-point increase in dissimilarity. The Black estimates are less precise in the sibling samples, which is expected given that the sibling subsample contains 12,314 Black individuals compared with 125,075 in the IV sample, but the point estimates sit within the confidence intervals of the full-sample estimates. Whatever variation in sample composition the identification strategies introduce, it does not by itself move the association.

Two features of these associations deserve emphasis before turning to the causal estimates. First, the coefficient is negative for *both* groups. This is the pattern that Panel (A) of Figure 1 describes

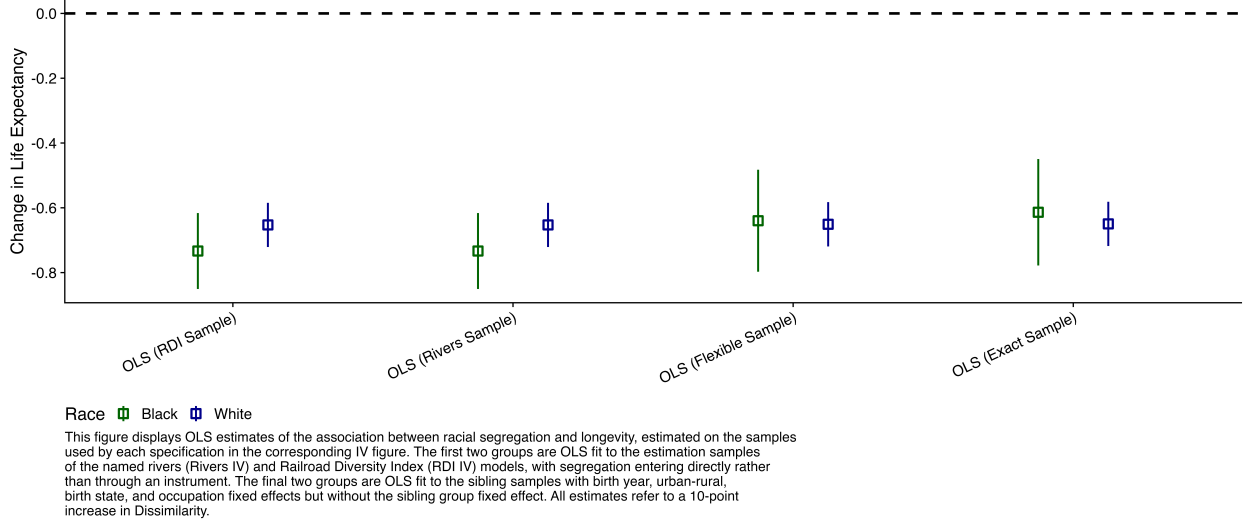


Figure 3: OLS Estimates of the Association Between Segregation and Longevity, by Race and Estimation Sample

and is inconsistent with the zero-sum expectation in Panel (B), which requires a positive coefficient for Whites. Second, the Black coefficient is larger in magnitude than the White coefficient, so the difference between them, which is the implied contribution of segregation to the Black-White longevity gap, is roughly 0.06–0.08 years per 10-point increase in dissimilarity. A Black-White gap therefore emerges here not because Whites gain from segregation but because Blacks lose more from it than Whites do. These associations are nevertheless descriptive. Individuals sort into counties for reasons plausibly related to their health, so I turn to the instrumental variable estimates to recover the causal effect.

8.2 Instrumental Variable Estimates

Tables 4 and 5 report the two-stage least squares estimates using the railroad and rivers instruments, and Figure 4 displays the estimates from all four identification strategies side by side. Consistent with the OLS associations, segregation reduces the longevity of both Black and White Americans, and every estimate in Figure 4 lies below zero.

Using the RDI instrument in the fully specified models, a 10-point increase in dissimilarity reduces Black longevity by 0.62 years (95% CI: $-0.75, -0.48$) and White longevity by 0.53 years (95% CI: $-0.66, -0.41$). The rivers instrument, which relies on natural rather than human-placed boundaries, yields estimates of 0.33 fewer years for Black individuals (95% CI: $-0.51, -0.16$) and

0.37 fewer years for White individuals (95% CI: -0.41 , -0.33). The two strategies bracket a common conclusion. A 10-point increase in segregation costs Black Americans between roughly four and seven months of life and White Americans between roughly four and six months, and in no specification is the White estimate positive or statistically indistinguishable from zero.

The sign pattern is the central result. The zero-sum account summarized in Panel (B) of Figure 1 requires that the coefficient on segregation be negative for Blacks and *positive* for Whites. That pattern does not appear in any specification, in either instrument strategy, or in either sibling design. Nor do the estimates support the Black-specific harm account in Panel (C), which requires a null effect for Whites; the White estimates are negative and precisely estimated, with confidence intervals that exclude zero by a wide margin in every case. The results are instead consistent with Panel (A): segregation is harmful to both groups, and the Black-White gap arises because the harm is unequal in size rather than because it is opposite in sign.

Reading the gap off the levels makes this concrete. Under the RDI estimates, a 10-point increase in dissimilarity widens the Black-White longevity gap by 0.09 years, but it does so while lowering the longevity of both groups. The gap and the levels tell different stories about the same coefficients: an analysis that examined only the gap would recover a Black disadvantage of about a month per 10 points of segregation and would be silent on the fact that both groups lose roughly half a year. This is precisely the ambiguity that Figure 1 describes, and it is why the group-specific estimates rather than the difference between them carry the substantive interpretation.

The magnitude of these effects is substantial when placed against other predictors of longevity in these data. In the fully specified OLS models, one additional year of education is associated with 0.04 additional years of life for Black individuals and 0.055 for White individuals. A 10-point increase in segregation therefore costs Black individuals the longevity equivalent of roughly 15 years of additional schooling and White individuals the equivalent of roughly 10 years. Put differently, the effect of a 10-point increase in dissimilarity is comparable in magnitude to about two-thirds of the estimated penalty of being male in these cohorts (-1.00 years for Blacks, -0.93 for Whites). Segregation is not a marginal input into how long these cohorts lived.

8.3 Triangulation Across Identification Strategies

Figure 4 assembles the estimates from all four strategies. The value of this comparison is that each design fails in a different way. The IV designs recover a local average treatment effect among compliers and depend on an exclusion restriction that cannot be tested directly; the sibling fixed effects design requires no exclusion restriction but identifies the effect only from siblings who ended up in counties with different levels of segregation, and does so on a much smaller and potentially selective sample. If the estimates converge despite these differences, it is unlikely that any single design's assumptions are generating the result.

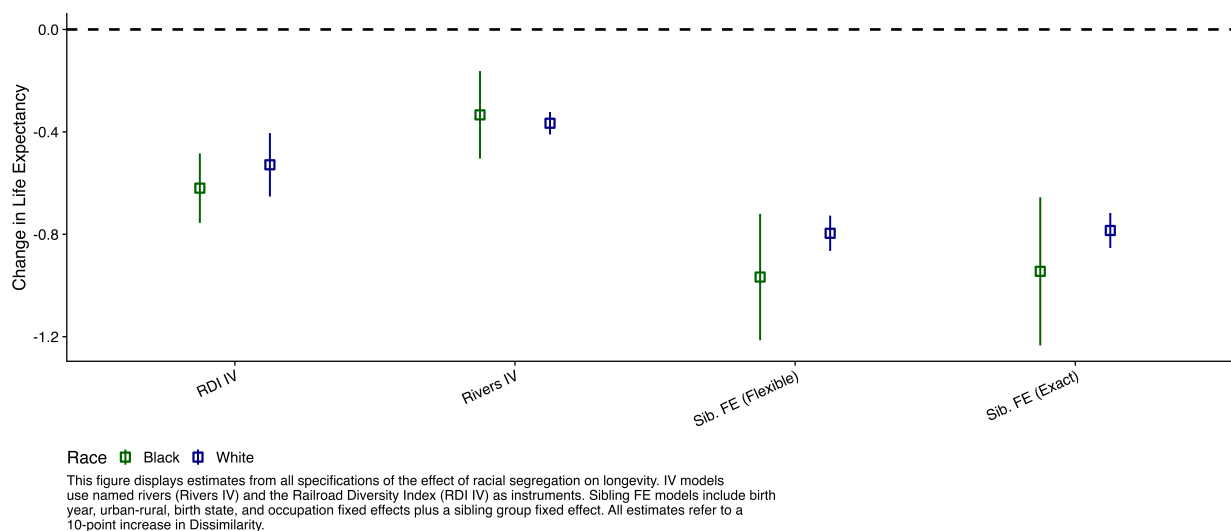


Figure 4: Estimates of the Effect of Segregation on Longevity Across Identification Strategies, by Race

They converge. All eight estimates in Figure 4 are negative and statistically distinguishable from zero, and they span a range of roughly 0.33 to 0.95 fewer years of life per 10-point increase in dissimilarity. The sibling fixed effects estimates are the largest in magnitude: 0.95 fewer years for Black individuals and 0.80 for White individuals under the flexible match, with the exact match yielding nearly identical values of 0.94 and 0.79. That the two sibling matching procedures agree so closely suggests that the estimates are not sensitive to how siblings are identified. The IV estimates are the most conservative, and the OLS associations in Figure 3 sit between the two IV strategies. Notably, the sibling estimates exceed the OLS associations estimated on the same samples, which is the pattern expected if families that sort into more segregated counties are positively selected on

unobserved determinants of longevity, such that failing to difference out family-level heterogeneity biases the association toward zero.

Across every design, the ordering of the two groups is preserved: the Black estimate is larger in magnitude than the White estimate in three of the four strategies, and in the rivers specification the two are statistically indistinguishable from one another. What does not vary across designs is the sign. Whether segregation is instrumented by nineteenth-century railroad configuration, by the arrangement of natural waterways, or differenced out at the family level, more segregation means shorter lives for Black and White Americans alike.

8.4 Heterogeneity by education and gender

- Whites: harm declines with education but never vanishes (college+ still harmed). *[[supports H3]]*
 - Blacks: harm roughly education-invariant. *[[supports H3 + differential-returns lit]]*
 - Tie explicitly back to the DuBois-derived H3, not just reported as a subgroup split. *[[closes the loop RA flagged on front/back-end connection]]*
- [[Keep Gender but compress — it's secondary.]]*

8.5 Mechanism: political-economy mediation

[[NEW headline subsection — B_RE3 payoff.]]

- Show segregation → policy mediators → longevity.
 - Report the mediated share of the total effect.
 - Interpret: supports the macro/shared channel (H2) as a real pathway, not just a residual.
- [[C_RI3 — mediated account, stated carefully]]*

To test the importance of several political-economic mechanisms linking segregation to longevity, I conduct a stepwise analysis of mechanisms compared to the baseline IV regressions (equation 4). I start with estimates using the RDI stratified by race and include

8.6 Double Truncation and Mortality Crossover

Thus far the estimates presented suggest that segregation had significant, negative effects on the longevity of both Black and White Americans aged 65 and older. However, because of this selective period in life, this raises the possibility that race-specific processes of mortality selection could bias these estimates. Indeed, a large literature in demography focused on the so-called Black-White mortality crossover has found that advanced ages, positively-selected Black older adults have lower mortality risk compared to White older adults of the same age. In this setting, it could be perfectly feasible that the magnitude of the negative estimates for Whites could be accounted for by this selection and the magnitude of the estimates for Blacks could be a significant underestimate of the full magnitude. I present two pieces of evidence against this conclusion.

Literature on the Black-White mortality cross-over generally finds that the crossover occurs at advanced ages. Prior scholarship has generally found that the crossover occurs between the mid-to-late 80s (see (?) for a review) while the majority of this sample is between the ages of 67-82, bias from the cross-over only plausibly affects the oldest members in the sample. To provide a quantification of the bias induced by truncated data, (Goldstein et al., 2023) developed a Gompertz maximum likelihood procedure to aid in the modeling of partially observed cohorts. The Gompertz procedure takes into account the truncated nature of the data and re-estimates the associations recovered by a traditional regression model using a log of the Gompertz distribution to correct estimates. This allows for a more accurate estimation of longevity and characterization of group-differences in longevity, which allows a quantification of the potential for the mortality cross-over to bias estimates directly.

Figure X provides a graphical representation of my main regression estimates by race in comparison to the Gompertz-corrected estimates. The Gompertz estimates are obtained by modeling death age as a function of segregation, birth year, birth state, and urban-rural fixed effects. This estimate is then paired with an identical OLS regression of the same form for a direct comparison of the bias between estimates. Results reveal that the bias induced by double truncation is minimal in magnitude and results reach the same conclusions as the main estimates. The Gompertz-corrected estimate for Black longevity is 12 percent larger than the regression-based estimate and the White longevity estimate is 6 percent larger. This increase in the magnitude is consistent with past find-

ings that estimates with truncated data are biased toward zero. These corrected estimates suggest that the Black-White longevity gap estimated via regression methods is slightly attenuated from the true estimate of this difference as evidenced by the larger difference in coefficients in the Gompertz estimate.

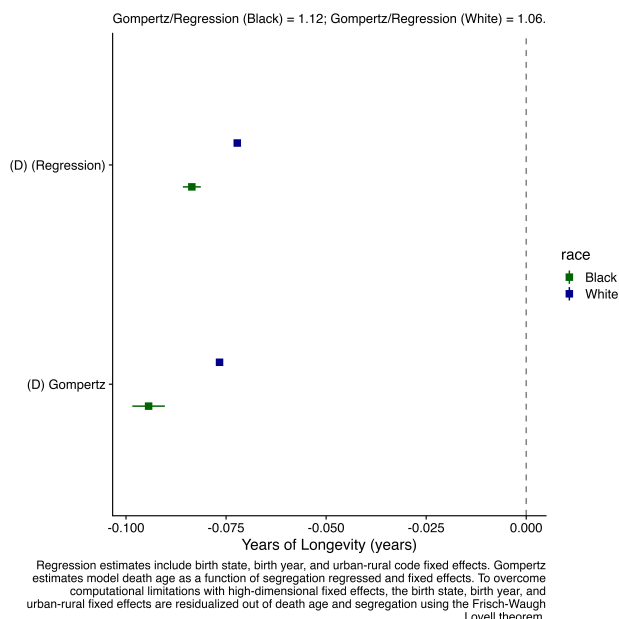


Figure 5: Double-Truncation-Corrected Estimates of the Relationship Between Segregation and Longevity

8.7 Military and cohort variation

9 Discussion

1. **Restate the verdict in level terms.** Segregation lowers longevity for both groups; the zero-sum (H1) sign pattern is rejected. *[[the whole reframe lands here]]*
2. **Why both groups lose: the macro channel.** The mediation results show the political-economy pathway is doing real work — this is what neighborhood-only accounts miss. *[[C_RA3 / the non-zero-sum case, now empirically backed]]*
3. **Class gradient among Whites.** Connect to wages-of-Whiteness: the White “benefit” is symbolic, the harm material; least-buffered (low-ed) Whites lose most. *[[C_RI4 payoff]]*

4. **Relation to prior work** (Logan–Parman, Karbeah–Hacker, Hendi), now characterized accurately and in level terms. *[[C-RA4]]*
5. **Limitations** (tighten, honest): survival-to-65 selection + crossover; historical cohorts vs. contemporary attenuation; LATE/exclusion caveats; mediation is partial, not exhaustive. *[[RE.1, RE.2; exclusion caveats]]*
6. **Implications.** Desegregation could raise longevity for both groups and is not a White-cost policy. Future work: contemporary cohorts, allostatic load, direct mechanism studies.

10 Conclusion

[[Short. One paragraph. The takeaway: racial stratification harms Black AND White Americans; the zero-sum belief is a symbolic mechanism, not a description of material reality; the harm partly flows through the political economy of segregated places.]]

Tables & Figures (planned)

[[Reuse most existing exhibits; new/changed ones marked NEW/CHANGED.]]

1. Fig 1 — Zero-sum processes
2. Fig 2 — OLS/multilevel level estimates by race
3. Fig 3 — IV level estimates by race. *[[CHANGED — level framing]]*
4. Fig 4 — **Triangulation panel:** all instruments + sibling FE. *[[NEW — lead figure]]*
5. Fig 5 — IV by education, Whites. (keep)
6. Fig 6 — IV by education, Blacks. (keep)
7. Fig 7 — **Mediation: segregation → policy → longevity.** *[[NEW]]*
8. Table 1 — Representativeness (+ race-specific match quality). *[[CHANGED]]*
9. Table 2 — Descriptives by race. (keep)

Table 2: Sample Filtering Criteria

Sample	RDI IV		Rivers IV	
	N (Counties)	N (Persons)	N (Counties)	N (Persons)
All Counties	3125	2613272	3125	2613272
Counties With >0 NH Black Pop	3091	2605102	3091	2605102
Counties With non-missing Instrument	3067	2601884	3067	2601884
Filter non-Missing Individual-and County-Level Variables	2955	2412213	2955	2412213

Individual-level characteristics include demographics, education, and fixed effects. Individuals are also filtered for missing post-stratification weights. The instrument filter is non-missing RDI and railroad density in the RDI columns and non-missing named rivers and stream density in the rivers columns.

10. Table 3 — OLS/multilevel results. *[[CHANGED]]*
11. Table 4 — IV results (level, by race). *[[CHANGED]]*
12. Table 5 — IV by gender. (keep)
13. Table 6 — IV by education. (keep)
14. Table 7 — **Mediation / MSEM results.** *[[NEW]]*

11 Appendix

11.1 A

1. Table 1 — filtering table;
2. Table 2 — F-stats
3. Table 3 —
4. Table — weighted models
5. Table — alt-index models;
6. Table — migration check;
7. Table — map

11.2 B: Robustness checks

11.3 Monotonicity

It is well known that segregation levels have declined over time. If part of that decline reflects less economic and racial residential sorting, then we would expect the association between physical division and segregation to weaken year-over-year. Following [Rauscher \(2020\)](#), evidence for monotonicity would suggest that the instrument strength declines over decades. Consistent with this assumption, the strength of the association between the instruments and segregation, measured by R^2 , decreases between 1980 and 2000. While this does not represent a conclusive test, these patterns are consistent with monotonicity. *[[Compress the robustness prose into a paragraph + appendix pointer: weights, alt indices, migration, cohort bins all concordant.]]*

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